

Criterion III.1

Live Presentation of Proposed Solution, System, Equipment and Technology

GOLDBOD TAS

Criterion III.1 Response

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Live Presentation of Proposed Solution, System, Equipment and Technology

Overview

Smart Innovations Ghana Limited presents a fully deployed, production-accessible web application for gold traceability and chain-of-custody management. The system is live, authenticated, and operational with real transaction data. It is not a prototype or mockup — it is a running platform with a populated stakeholder database, active licence registry, and interactive traceability graph built on real gold movement records.

Deployed System Architecture

Frontend	React.js single-page application. Responsive layout, collapsible navigation, role-based view rendering. Deployed and accessible via secure HTTPS.
Backend	Node.js / TypeScript REST API. Stateless, horizontally scalable. JWT-authenticated endpoints with role-based access control.
Database	PostgreSQL — relational store for actor registry, licence records, transaction history, and traceability events.
Traceability Engine	React Flow — interactive, zoomable directed graph rendering full chain-of-custody from source transaction through aggregation and batching to distribution.
Deployment	VPS-hosted, Nginx reverse proxy, PM2 process management, SSL/TLS via Certbot. CI/CD via GitHub Actions.
Source Repository	github.com/Tmony117/gold_Tracability_app — full source code accessible to GoldBod for review and audit.

Actor Registry — Live Modules

The system maintains a live registry of all licensed participants across Ghana's gold value chain. The following 11 actor categories are registered, managed, and actively tracked in the deployed platform:

Small Scale Miners	Licensed ASM operators — registration, status, concession zone
Buyer License Tier 1	Community-level gold buyers — licence issuance and expiry tracking
Buyer License Tier 2	Regional aggregating buyers — higher volume, stricter compliance
Aggregator License	Multi-source aggregation operators — batch consolidation
Self-Financing Aggregators	Aggregators operating with independent capital
Refinery License	Formal gold refinery operators
Smelting License	Smelting facility operators (kitchen-equivalent per TOR)
Importer / Fabricator	Gold import and fabrication licence management

Storage License	Licensed storage facility operators
Transporter License	Licensed gold transporters — custody-in-transit tracking
Assay Centers	Certified assay laboratory management

Screenshot 1 — Live Dashboard

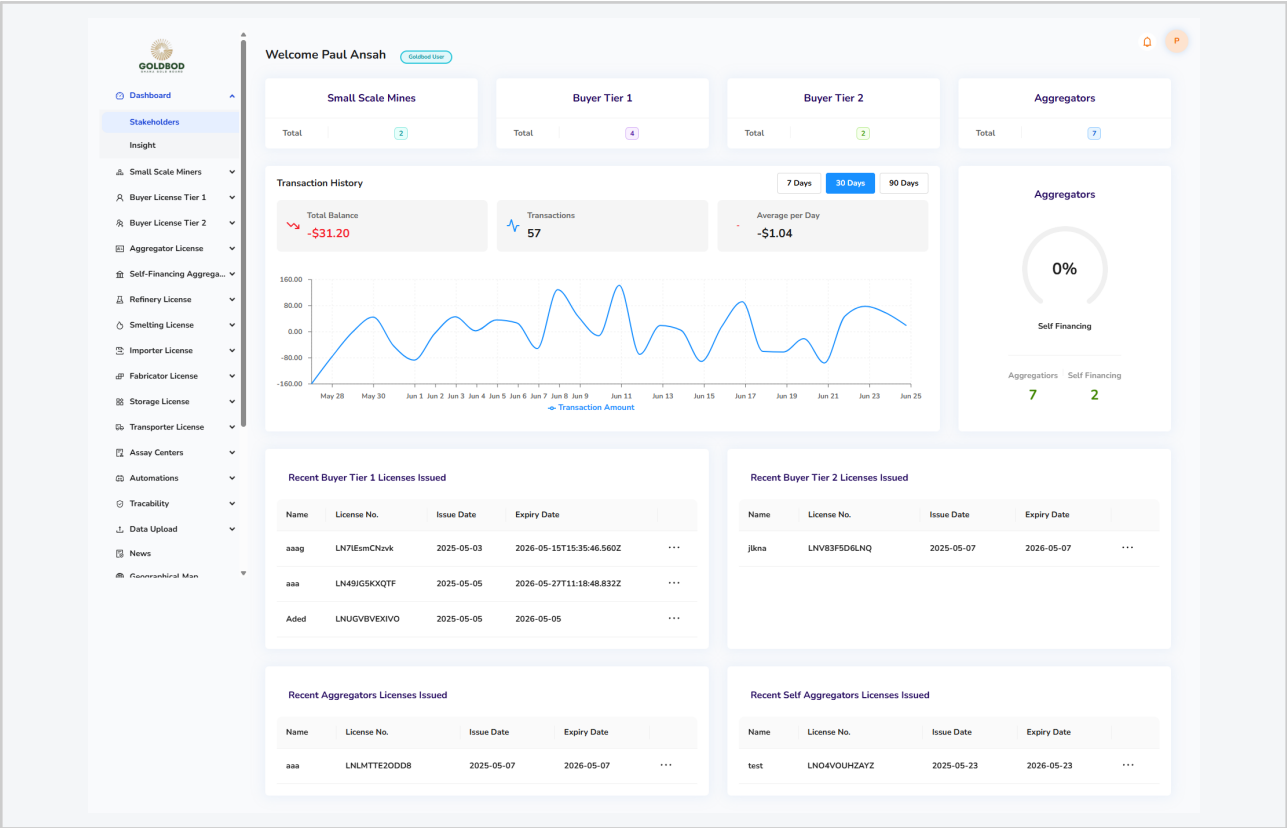


Figure 1 — GoldBod TAS Live Dashboard (authenticated session: Paul Ansah, Goldbod User). Real-time actor counts: 2 Small Scale Mines, 4 Buyer Tier 1, 2 Buyer Tier 2, 7 Aggregators. 30-day transaction history chart with 57 transactions. Recent licence tables across all actor types.

Screenshot 2 — Gold Traceability Graph

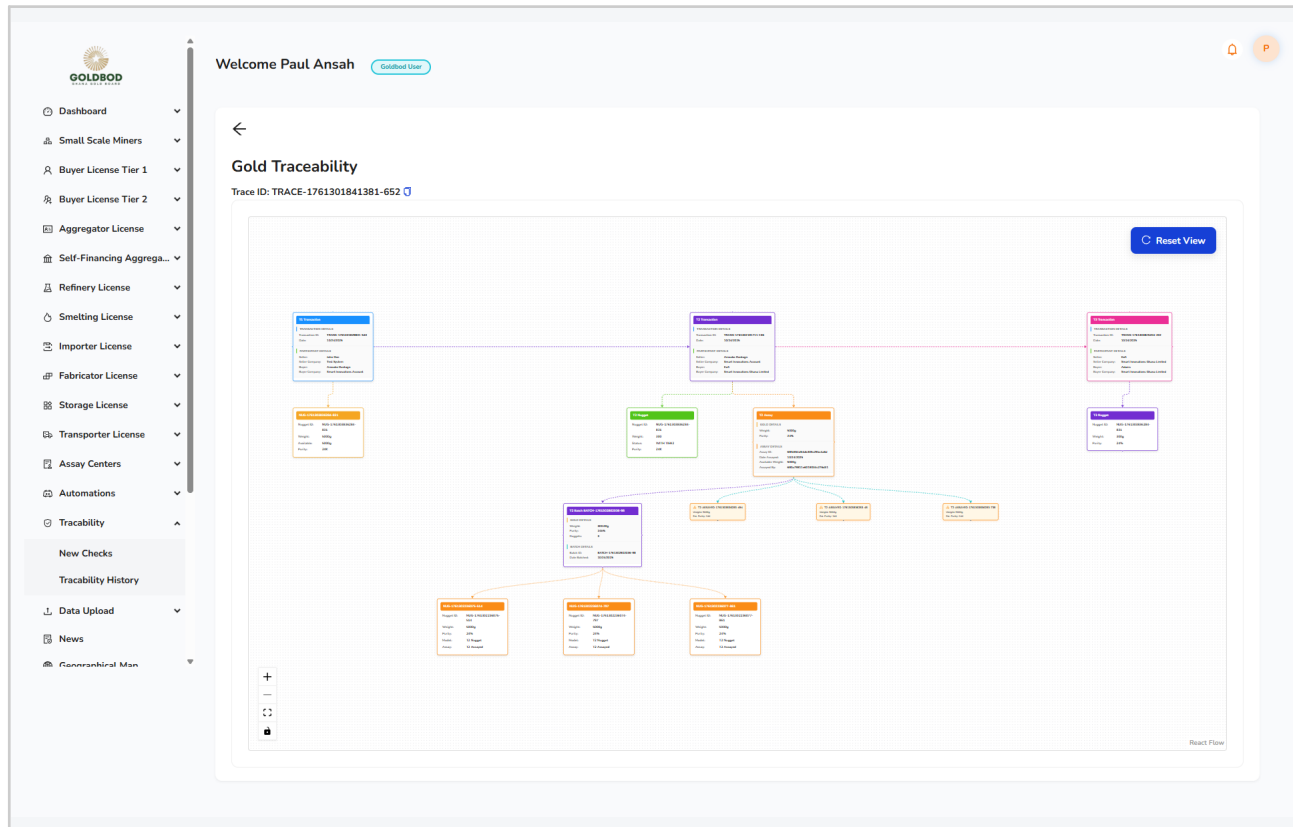


Figure 2 — Interactive Gold Traceability Graph (Trace ID: TRACE-1761301841381-652). React Flow directed graph: source transactions (blue) → Tier 6 Nuggets (orange/green) → Batch consolidation (purple) → distributed nuggets. Each node shows weight, purity, provenance.

EVIDENCE PROVIDED

- ✓ Live deployed web application with authenticated HTTPS access — verifiable at submission
- ✓ Full actor registry: all 11 licence categories defined in GoldBod TOR Section 2.2.3
- ✓ Interactive chain-of-custody graph with unique Trace ID referencing (React Flow)
- ✓ 30-day transaction history: 57 live transactions recorded on deployed system
- ✓ Full gold value chain covered: miner → Tier 1/2 buyer → aggregator → smelting → GoldBod
- ✓ Source code repository: github.com/Tmony117/gold_Tracability_app